

DARSHAN RANA

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PROFILE SUMMARY

Results-driven technology professional with over 2+ years of experience in Manufacturing IT, MES applications, and **Data Engineering**. Proficient in developing automation solutions using Python, optimizing **SQL queries** for large-scale production databases, and managing **ETL pipelines** to ensure reliable data flow. Skilled at delivering real-time insights for decision-making in manufacturing environments, with proven success at Tata Motors. Experienced in collaborating with production, quality, and IT teams to implement robust data systems that improve efficiency and reduce downtime. Currently advancing expertise in **Big Data frameworks, PySpark, Databricks, and AWS Cloud services** to design scalable, high-performance **data solutions** for enterprise-level applications.

WORK EXPERIENCE

Data Engineer & MES Developer – Tata Technologies (Primus Global Technologies Pvt. Ltd.)

| Ahmedabad | May 2026 – Current

- Collaborated with MES department (Apriso) at Tata Motors Passenger Vehicle Plant, Sanand, delivering IT-enabled solutions for manufacturing operations.
- Spearheading core **Data Engineering** workflows and MES application enhancements for the passenger vehicle manufacturing lines.
- Engineered and optimized complex **SQL queries, stored procedures, and ETL pipelines** to handle real-time, high-volume production data.
- Leading backend database management, troubleshooting data flow integrity, and scaling data architecture for shop-floor automation systems.

Databricks Developer – LinguistHubs | Remote | July 2025 – Feb 2026

- Designed and implemented a scalable cloud data platform using Databricks to process, store, and analyze large-scale structured and unstructured linguistics datasets.
- Developed robust PySpark ETL workflows to ingest massive volumes of text data, applying complex data cleansing and transformation logic.
- Implemented Delta Lake for ACID compliance and time travel capabilities, ensuring high data reliability across distributed environments.
- Optimized Databricks cluster configurations and Spark SQL queries, significantly enhancing query performance and reducing analytical response times by over 40%.

Data Consultant – Tata Technologies (Vega Intellisoft Pvt. Ltd.) | Ahmedabad | Feb 2024 – June 2025

- Collaborated with MES department (FTPC) at **Tata Passenger Electric Mobility Plant**, Sanand, delivering IT-enabled solutions for manufacturing operations.
- Developed Python-based shop-floor forms automated manual data entry, ensuring accurate real-time capture of production data. Designed and optimized **SQL stored procedures, triggers, and ETL jobs** to manage high-volume production data efficiently.
- Created and maintained SparkPlug Portal **SQL queries** that enabled live tracking of production line activities for Tata Motors employees. Implemented **SQL-based data validation frameworks** to eliminate duplicate and inconsistent records, significantly improving reporting accuracy.
- Streamlined **ETL processes** in Microsoft **SQL Server**, enhancing performance and reducing processing time for large datasets. Provided MES application support, troubleshooting real-time production issues, minimizing downtime, and improving system reliability.
- Collaborated with cross-functional teams (production, quality, IT) to align technical solutions with business requirements.

- Completed intensive training in Python-based **data analysis** and visualization using Pandas, NumPy, Seaborn, and Matplotlib libraries.
- Performed preprocessing, exploratory data analysis (EDA), and statistical evaluation on large datasets to generate business insights by creating charts in Microsoft Power BI and Jupyter Notebook.
- Delivered **data analysis projects** that demonstrate practical applications of machine learning and visualization techniques by Microsoft Power BI.

PROJECTS

SparkPlug - Tata Passenger Electric Mobility Plant

- Architected automated **ETL pipelines** using **Databricks and PySpark** to process high-volume manufacturing data from SQL Server into a cloud data lake.
- Engineered **Delta Lake** architecture to manage real-time streaming and batch processing of shop-floor data for the SparkPlug portal.
- Optimized complex data transformations and distributed queries using **Spark SQL**, accelerating the generation of daily, monthly, and yearly plant-wide production reports.
- Collaborated closely with the Data Engineering team to optimize query performance, ensuring high data quality and accurate, real-time data availability for key stakeholders.

LinguistHubs Cloud Database Development - <https://linguisthubs.com>

- Designed and implemented a scalable cloud data platform using **Databricks** to process, store, and analyze large-scale structured and unstructured linguistics datasets.
- Developed robust **PySpark ETL workflows** to ingest massive volumes of text data, applying complex data cleansing and transformation logic.
- Implemented **Delta Lake** for ACID compliance and time travel capabilities, ensuring high data reliability across distributed environments.
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Recall Phase 2 - Tata Passenger Electric Mobility Plant

- Developed and deployed shop operation forms using Rockwell Optix Automation to capture accurate production data in real time.
- Configured **complex SQL logic** to detect and prevent duplicate entries at the database level, **reducing data discrepancies by 95%**.
- Designed error alert mechanisms to notify operators instantly about invalid inputs, minimizing data corruption on the shop floor.
- Successfully integrated solution into live production lines, improving reporting accuracy and operational efficiency.

E-Radar Monitoring System - Tata Passenger Electric Mobility Plant

- Engineered a **real-time data monitoring system** to track data flow integrity across multiple checkpoints in the production line.
- Integrated Rockwell Optix Automation with **SQL Server** to identify and log missing data entries, improving traceability of production events.
- Developed timestamp-based missing data detection, enabling swift troubleshooting and **reducing resolution time by 30%**.

EDUCATION

Bachelor of Engineering (IT) | Sardar Patel College of Engineering | 2019 – 2023

- Gujarat Technological University | CGPA: 8.22

Higher Secondary School (Science Stream) | Shree S.Z. Waghela Highschool | 2017 – 2019

- GHSEB, Khambhat | Percentage: 71%

CERTIFICATIONS

- **SQL(Advance)** - HackerRank
- **AWS Cloud Essentials** - Amazon Web Services
- **Azure AI** - Microsoft & Edunet Foundation
- **Apache Spark** - Databricks
- **Databricks Course** - Databricks
- **Data Engineering Course** - IBM

SKILLS

- **Programming:** Python (**Data Processing, ETL, Automation**), **SQL, PostgreSQL, T-SQL, PySpark, FTPC**
- **Databases:** Microsoft **SQL Server**, MongoDB, MySQL, Delta Lake
- **Data Analytics & Visualization:** Databricks, EDA, Microsoft Excel, Power BI, **Spark SQL, SQL Query Optimization**
- **Big Data & Cloud:** AWS, Azure, PySpark, Databricks
- **Version Control:** Git, GitHub
- **Soft Skills:** Adaptability, Continuous Learning, Positive Attitude, Teamwork, Time Management

REFERENCES

- **Pawan Khade** (Senior Developer, Capgemini) | Phone: +91 8605254271 | Email: khade36@gmail.com
- **Vivek Bind** (Manager, Reliance Solar) | Phone: +91 9795813378 | Email: vivek.bind@ril.com